EE2005: Electrical Machines
Tutorial 10

1. A 2 - pole, 50 Hz induction motor supplies 15 kW to a load at a speed of 2950 rpm.

   (a) What is the motors slip?

   (b) What is the developed torque in the motor in Nm under these conditions?

   (c) What will the operating speed of the motor be if its torque is doubled?

   (d) How much power will be supplied by the motor when the torque is doubled?

   [**Ans:** (a) 1.67% (b) (b) 48.6 Nm (c) 2900 rpm (d) 29.5 kW]

2. A 460 V, 25 hp, 60 Hz, 4-pole, Y-connected induction motor has the following impedances in ohms per phase referred to the stator circuit:
   $R_1 = 0.641\,\Omega$, $R_2' = 0.332\,\Omega$, $X_1 = 1.106\,\Omega$, $X_2' = 0.464\,\Omega$, $X_m = 26.3\,\Omega$
   The total rotational losses are 1100 W and are assumed to be constant. The core loss is lumped in with the rotational losses. Consider the induction motor in Q.2 ,

   (a) What is the maximum torque of this motor? At what speed and slip does it occur?

   (b) What is the starting torque of this motor?

   (C) When the rotor resistance is doubled, what is the speed at which the maximum torque now occurs? What is the new starting torque of the motor?

   [**Ans:** (a) 240.185 Nm, 1447.56 rpm, slip = 0.1958 (b) 109.24 Nm (c) 1095.12 rpm, 178.83 Nm]

3. A 460 V, 1740 rpm, 60 Hz, 4-pole, wound rotor induction motor has the following parameters per phase:
   $R_1 = 0.25\,\Omega$, $R_2' = 0.2\,\Omega$, $X_1 = 0.5\,\Omega$, $X_2' = 0.5\,\Omega$, $X_m = 30\,\Omega$
   The rotational losses are 1700 W. With the rotor terminals short-circuited, find

   (a) (i) Starting current when started direct on full voltage.
   (ii) Starting torque.

   ((b) (i) Full-load slip.
   (ii) Full-load current.
   (iii) Ratio of starting current to full-load current.
   (iv) Full-load power factor.

   Assume approximate equivalent circuit.

   [**Ans:** (a) (i)250.29 $\angle$ -66.66° A (ii) 186.7 Nm (b) (i) 0.03333 (ii)43.83 $\angle$ -20.51° A (iii) 4.259 (iv) 0.93 lag]

4. A three-phase, 460 V, 60 Hz, six-pole wound-rotor induction motor drives a constant load of 100 Nm at a speed of 1140 rpm when the rotor terminals are short-circuited. It is required to reduce the speed of the motor to 1000 rpm by inserting resistances in the rotor circuit. Determine the value of the resistance if the rotor winding resistance per phase is 0.2 Ω. Neglect rotational losses. The stator-to-rotor turns ratio is unity.

   [**Ans:** 0.468 Ω per phase]

5. A 6-pole, 50 Hz, 3-phase induction motor running on full load develops a useful torque of 162 Nm, and it is observed that the rotor emf makes 90 complete cycles per min. Calculate the power output. If the mechanical torque lost in friction be 13.5 Nm, find the copper loss in rotor windings, the input to the motor and efficiency. The stator losses are 750 W.

   [**Ans:** Rotor copper loss = 551.35 W, Efficiency = 86.02 %]

6. A 400 V, 50 Hz, 30 hp, 3-ϕ induction motor is drawing 50 A current at 0.8 power factor lagging. The stator and rotor copper losses are 1.5 kW and 900 W respectively. The friction and windage losses are 1050 W and the core losses are 1200 W. Calculate the air-gap power of the motor.

   [**Ans:** 25 012.8 W]

7. The blocked rotor test on a 37 kW, 440 V, 50 Hz, 6-pole, Y-connected, 3-ϕ induction motor gave the following results: $V_{line} = 230$ V , $I_{line} = 140$ A, $\cos\phi = 0.35$. The motor drives a load having a constant torque of 300 Nm. Estimate the maximum percentage reduction in supply voltage possible before the motor stalls. Assume that the copper losses are equally divided between the stator and the rotor. Neglect magnetising current.

   [**Ans:** 41.1 %]

8. An 8-pole, 50 Hz, 3-ϕ induction motor has an equivalent rotor resistance of 0.07 Ω/phase. If its stalling speed is 630 rpm, how much resistance is to be included per phase to obtain the maximum torque at starting? Ignore magnetising current.

   [**Ans:** 0.3675 Ω]

9. An induction motor has an effiency of 0.9 when the load is 50 hp. At this load, the stator and rotor copper loss each equal the iron loss. The mechanical losses are one-third of the no-load loss. Find the slip.

   [**Ans:** 0.0303]

10. A 400 V, 50 Hz, 4-pole, 1400 rpm, Y-connected squirrel cage induction motor has the following parameters referred to the stator: $R_2' = 1.0\,\Omega$, $X_1 = X_2' = 1.5\,\Omega$. Neglect stator resistance and core and rotational losses of the motor. The motor is controlled from a 3-phase voltage source inverter with constant V/f control. Calculate the stator voltage and frequency to obtain the maximum torque at starting. Use the approximate equivalent circuit.

    [**Ans:** 16.7 Hz, 133.33 V]